Sample Means Hypothesis Testing

Questions 1–20

Question 1. In each of these situations, find the p-value and hence determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \mu = 30$, $H_1 : \mu < 30$
$X \sim N(30, 2^2)$
$n = 4$, $\bar{x} = 27.1$, 5% significance level

b) $H_0 : \mu = 45$, $H_1 : \mu < 45$
$X \sim N(45, 16)$
$n = 10$, $\bar{x} = 43.2$, 5% significance level

c) $H_0 : \mu = 50$, $H_1 : \mu < 50$
$X \sim N(50, 20)$
$n = 30$, $\bar{x} = 49.2$, 10% significance level

d) $H_0 : \mu = 85$, $H_1 : \mu < 85$
$X \sim N(85, 37)$
$n = 15$, $\bar{x} = 81.2$, 1% significance level

Question 2. In each of these situations, find the p-value and hence determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \mu = 60$, $H_1 : \mu > 60$
$X \sim N(60, 3^2)$
$n = 5$, $\bar{x} = 62.1$, 5% significance level

b) $H_0 : \mu = 23$, $H_1 : \mu > 23$
$X \sim N(23, 25)$
$n = 9$, $\bar{x} = 25.9$, 5% significance level

c) $H_0 : \mu = 104$, $H_1 : \mu > 104$
$X \sim N(104, 35)$
$n = 20$, $\bar{x} = 107$, 10% significance level

d) $H_0 : \mu = 162$, $H_1 : \mu > 162$
$X \sim N(162, 54)$
$n = 25$, $\bar{x} = 166.3$, 1% significance level

Question 3. In each of these situations, find the p-value and hence determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \mu = 34$, $H_1 : \mu \neq 34$
$X \sim N(34, 6^2)$
$n = 16$, $\bar{x} = 30$, 5% significance level

b) $H_0 : \mu = 70$, $H_1 : \mu \neq 70$
$X \sim N(70, 2.25)$
$n = 7$, $\bar{x} = 70.8$, 5% significance level

c) $H_0 : \mu = 125$, $H_1 : \mu \neq 125$
$X \sim N(125, 80)$
$n = 100$, $\bar{x} = 126.6$, 10% significance level

d) $H_0 : \mu = 216$, $H_1 : \mu \neq 216$
$X \sim N(216, 17)$
$n = 30$, $\bar{x} = 214$, 1% significance level

Question 4. In each of these situations, find the critical region and hence determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \mu = 28$, $H_1 : \mu < 28$
$X \sim N(28, 2.5^2)$
$n = 18$, $\bar{x} = 26.7$, 5% significance level

b) $H_0 : \mu = 32$, $H_1 : \mu < 32$
$X \sim N(32, 25)$
$n = 26$, $\bar{x} = 30.5$, 5% significance level

c) $H_0 : \mu = 106$, $H_1 : \mu < 106$
$X \sim N(106, 60)$
$n = 33$, $\bar{x} = 104.4$, 10% significance level

d) $H_0 : \mu = 340$, $H_1 : \mu < 340$
$X \sim N(340, 129)$
$n = 80$, $\bar{x} = 336.6$, 1% significance level

Question 5. In each of these situations, find the critical region and hence determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \mu = 75$, $H_1 : \mu > 75$
$X \sim N(75, 10^2)$
$n = 19$, $\bar{x} = 79.3$, 5% significance level

b) $H_0 : \mu = 160$, $H_1 : \mu > 160$
$X \sim N(160, 121)$
$n = 33$, $\bar{x} = 162.7$, 5% significance level

c) $H_0 : \mu = 1250$, $H_1 : \mu > 1250$
$X \sim N(1250, 3.2)$
$n = 4$, $\bar{x} = 1252.3$, 10% significance level

d) $H_0 : \mu = 670$, $H_1 : \mu > 670$
$X \sim N(670, 163)$
$n = 60$, $\bar{x} = 674.1$, 1% significance level

Question 6. In each of these situations, find the critical region and hence determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \mu = 85$, $H_1 : \mu \neq 85$
$X \sim N(85, 7^2)$
$n = 54$, $\bar{x} = 82.9$, 5% significance level

b) $H_0 : \mu = 1.6$, $H_1 : \mu \neq 1.6$
$X \sim N(1.6, .04)$
$n = 43$, $\bar{x} = 1.68$, 5% significance level

c) $H_0 : \mu = 57$, $H_1 : \mu \neq 57$
$X \sim N(57, 29)$
$n = 10$, $\bar{x} = 54.04$, 10% significance level

d) $H_0 : \mu = 890$, $H_1 : \mu \neq 890$
$X \sim N(890, 250)$
$n = 9$, $\bar{x} = 902.9$, 1% significance level

Question 7. In each of these situations, find the test statistic and the critical value of Z, and hence determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \mu = 35$, $H_1 : \mu < 35$
$X \sim N(35, 8^2)$
$n = 6$, $\bar{x} = 29.7$, 5% significance level

b) $H_0 : \mu = 60$, $H_1 : \mu < 60$
$X \sim N(60, 50)$
$n = 10$, $\bar{x} = 55.9$, 5% significance level

c) $H_0 : \mu = 12.5$, $H_1 : \mu < 12.5$
$X \sim N(12.5, 1.4)$
$n = 4$, $\bar{x} = 11.9$, 10% significance level

d) $H_0 : \mu = 496$, $H_1 : \mu < 496$
$X \sim N(496, 244)$
$n = 45$, $\bar{x} = 490.1$, 1% significance level

Question 8. In each of these situations, find the test statistic and the critical value of Z, and hence determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \mu = 300$, $H_1 : \mu > 300$
$X \sim N(300, 20^2)$
$n = 3$, $\bar{x} = 319.3$, 5% significance level

b) $H_0 : \mu = 133$, $H_1 : \mu > 133$
$X \sim N(133, 1.21)$
$n = 5$, $\bar{x} = 133.7$, 5% significance level

c) $H_0 : \mu = 8.8$, $H_1 : \mu > 8.8$
$X \sim N(8.8, .56)$
$n = 12$, $\bar{x} = 9.085$, 10% significance level

d) $H_0 : \mu = 8550$, $H_1 : \mu > 8550$
$X \sim N(8550, 560)$
$n = 35$, $\bar{x} = 8560.3$, 1% significance level

Question 9. In each of these situations, find the test statistic and the critical value of Z, and hence determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \mu = 11$, $H_1 : \mu \neq 11$
$X \sim N(11, 1)$
$n = 8$, $\bar{x} = 10.25$, 5% significance level

b) $H_0 : \mu = 77$, $H_1 : \mu \neq 77$
$X \sim N(77, 35)$
$n = 22$, $\bar{x} = 78.9$, 5% significance level

c) $H_0 : \mu = 170$, $H_1 : \mu \neq 170$
$X \sim N(170, 560)$
$n = 6$, $\bar{x} = 154.8$, 10% significance level

d) $H_0 : \mu = 35600$, $H_1 : \mu \neq 35600$
$X \sim N(35600, 19500)$
$n = 85$, $\bar{x} = 35552$, 1% significance level

Question 10. Molly owns a 1950s inspired café with a fizzy drink machine that dispenses soda into cups to a volume V, where V can be modelled as normally distributed with a mean of 380 ml and standard deviation 3 ml. She believes that the machine is not working properly, and the amount of soda being dispensed is less than it should be. Molly carries out a hypothesis test, at the 10% significance level, to test her belief. She collects a random sample of 15 soda drinks and measures their mean volume to be 378.7 ml. Molly's hypothesis test is shown below.

Let X be the volume of soda dispensed by the machine in millilitres.
$H_0 : \mu = 380$
$H_1 : \mu > 380$
$\bar{X} \sim N\left(380, \frac{3^2}{15}\right)$
$P(\bar{X} < 378.7) = 0.0466$
$0.0466 < 0.1$ *so we fail to reject* H_0
There is insufficient evidence to suggest that the population mean amount of soda is less than expected.

a) Identify the first two errors made by Molly in her hypothesis test.
b) Write out a fully correct hypothesis test for Molly.

Question 11. The heights of a certain type of plant over the age of 2 years can be modelled by a normal distribution with a mean of 130 cm and a standard deviation of 20 cm. Trisha works at a garden centre that sells this type of plant, and believes that the mean height is actually greater than 130 cm. She collects a random sample of 10 of these plants and finds their mean height to be 138 cm. Carry out a test at the 3% significance level to determine whether there is evidence to support Trisha's belief.

Question 12. Between April 2017 and March 2018, the weekly household spending in the UK can be modelled by a normal distribution with a mean of £589.60 (adjusting for inflation to compare with April 2018 to March 2019), and a standard deviation of £120.30. A random sample of size 150 was taken from April 2018 to March 2019, which had a mean of £605. Carry out a hypothesis test, at the 10% significance level, to investigate whether the weekly household spending in the UK significantly changed from the 2017-2018 to 2018-2019 tax years.

Question 13. A confectionary company sells chocolate bars that are claimed to have a mean weight of 100 grams and a standard deviation of 5 grams. A quality control team is tasked with verifying if the mean weight of the chocolate bars is different from the claim. A random sample of 16 bars are selected from the production line, and their weights measured. The weights are shown in the table below.

97.7	97.1	97.2	97.2
98.8	97.6	100.2	96.5
96.5	97.0	96.5	96.9
98.4	99.7	100.0	99.5

Using a significance level of 5%, conduct a hypothesis test to determine if there is evidence to suggest that the mean weight of the chocolate bars produced by the company differs from the 100 grams claim.

Question 14. A bakery bakes loaves of bread. The mass of the loaves can be modelled by a normal distribution with a mean weight of 400 grams and a standard deviation of 6 grams. The length of time the loaves are in the oven is shortened and the bakery wants to test whether this will reduce the weight of the loaves. A large number of loaves are baked using this new method, and a random sample of 50 are selected and weighed. The mean weight for these 50 loaves is 398 grams. The bakery carry out a hypothesis test, at the 5% significance level, to test their belief. The bakery's hypothesis test is shown below.

Let X be the weight of baked loaves in grams

$H_0 : \mu = 400$

$H_1 : \mu \neq 400$

$\bar{X} \sim N\left(400, \frac{6^2}{\sqrt{50}}\right)$

$P(\bar{X} < a) = 0.05 \Rightarrow a = 396.3$

$398 > 396.3$ *so we fail to reject* H_0

There is insufficient evidence to suggest that the population mean weight of loaves of bread has been reduced.

a) Identify the first two errors made by the bakery in their hypothesis test.

b) Write out a fully correct hypothesis test for the bakery.

Question 15. The heights of female adults in the UK in 1998 can be modelled by a normal distribution with a mean of 161 cm and a standard deviation 9 cm. Gemma believes that the mean height of females in the UK is different to that in 1998. She decides to conduct a hypothesis test using a 5% significance level by collecting a random sample of 60 female adults.

a) Find the critical region for this test.

The mean height of the sample is 162.4 cm.

b) Write out and complete Gemma's hypothesis test.

Question 16. It is known that the waiting times at a veterinary surgery for those without an appointment can be modelled by a normal distribution with a mean of 45 minutes and a standard deviation of 6 minutes. A new manager, Dave, takes over the surgery and introduces some new processes that he hopes will reduce the waiting time for those without an appointment. Dave decides to conduct a hypothesis test at the 5% significance level to test his belief and will collect a random sample of how long 12 people with their pets wait.

a) Find the critical region for the test.

Dave collects the following data, where the times are in minutes.

41	39	49	46	38	58
33	42	36	38	45	44

b) Write out and complete Dave's hypothesis test.

Question 17. In 2019, a large study was undertaken to look into adults' sleeping habits, and it was found that the amount of time an adult sleeps can be modelled by a normal distribution with a mean of 7 hours and a standard deviation of 1.1 hours. In 2023, Stuart wanted to determine if there is any evidence to suggest that the average amount of sleep an adult gets has changed since 2019 and decides to carry out a hypothesis test at the 1% significance level to test his belief. Stuart collects a random sample of 80 adults and finds their average amount of sleep to be 7.3 hours. Stuart's hypothesis test is shown below.

Let X be the number of hours of sleep an adult gets.

$\mathrm{H}_0 : \mu = 7$

$\mathrm{H}_1 : \mu \neq 7$

$\bar{X} \sim \mathrm{N}\left(7.3, \frac{1.1^2}{80}\right)$

$Z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} = \frac{7.3-7}{\frac{1.1}{\sqrt{80}}} = 2.439$

$\mathrm{P}(Z > a) = 0.995 \Rightarrow a = 2.576$

$2.439 < 2.576$ *so we fail to reject* H_0

There is insufficient evidence to suggest that the population mean number of hours that adults sleep has changed.

a) Identify the first two errors made by Stuart in his hypothesis test.

b) Write out a fully correct hypothesis test for Stuart.

Question 18. A pharmaceutical company produces a cold and flu medication claiming that the mean time for their medication to relieve symptoms is 30 minutes, with a standard deviation of 6 minutes. Assume that these times are normally distributed. A team of medical researchers believes that it takes longer for the medication to take effect than the average time claimed. The team wishes to conduct a hypothesis test at the 1% significance level to determine if there is any evidence to suggest the mean time is longer than 30 minutes. A random sample of ten patients suffering with cold and flu symptoms is given the medication and their response times are recorded. The sample has a mean response time of 35 minutes.

a) Find the test statistic and the critical values of Z.

b) Write out and complete the test.

Question 19. A car manufacturer has just unveiled their latest model, and the manufacturer claims that the mean fuel efficiency is 40 miles to the gallon with a standard deviation of 2 miles to the gallon. An environmentalist watchdog wants to verify their claim and decides to conduct a hypothesis test at the 2% significance level to determine if there is evidence to suggest that the 40 miles to the gallon average is too high, and that actually it is less than the advertised figure. A random sample of 15 of the new cars are tested, and the sample has a mean fuel efficiency of 38.75 miles to the gallon.

a) Find the test statistic and the critical values of Z.
b) Write out and complete the test.

Question 20. An established fitness club owns several gyms across the country and has a large membership of adults aged 18 and over. All members who join the gym are enrolled on a twelve-week course that is focused on weight-loss and maintaining this level. The fitness club claim that the mean weight-loss for their membership is 8.5 kilograms, with a standard deviation of 1.8 kilograms. It may be assumed that the weight-loss for the membership is normally distributed. A group of nutritionists decide to investigate whether the mean weight loss achieved by the membership differs from 8.5 kilograms, and they decide to conduct a hypothesis test using a significance level of 3%. The nutritionists track a random sample of 15 members of the fitness club and the sample's mean weight loss is 9.2 kilograms. Write out and complete the nutritionists' hypothesis test.